AC GENERATION

Generator Construction

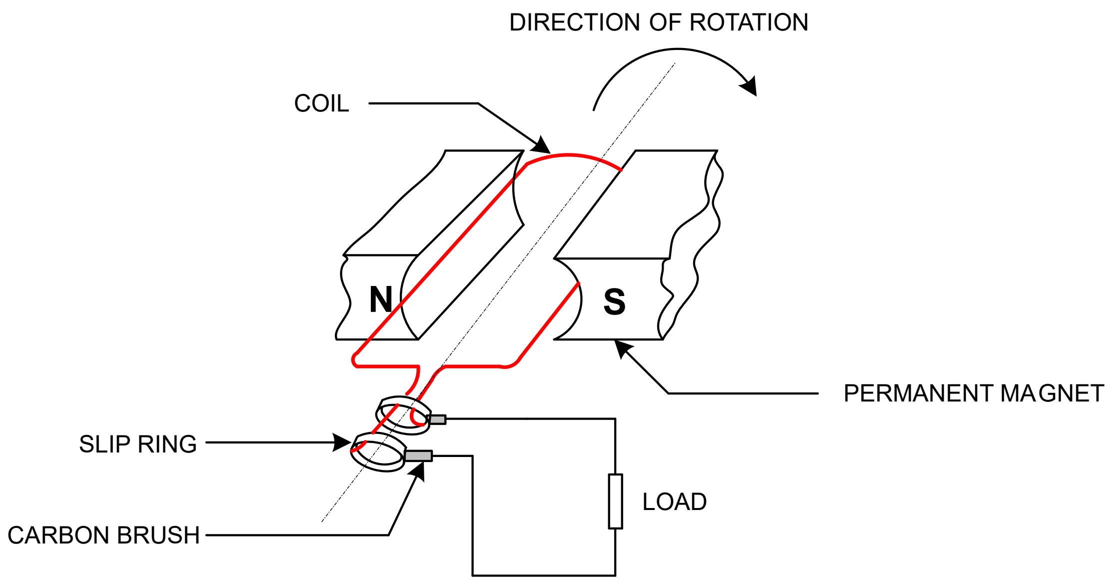

Fig 1.01 Simple AC generator

The generator consists of the following parts

(a) Permanent magnet of two poles that produces stationary magnetic field.

(b) A coil of conductor that is rotated in the magnetic field causing a voltage to be induced in it.

(c) Two slip rings connected to the two ends of the coil from which the generated voltage can be tapped and supplied to the load.

(d) Carbon brushes that are spring loaded attached to the slip rings that conduct current to the load.

(e) Conductors connected to the carbon brushes and the external load. They provide a path for current flow between the coil and the load.

PRINCIPLE OF OPERATION OF AN A.C. GENERATOR

The basic ac generator shown in fig.1.01 is a pivoted coil of sides AB and CD which rotates between the two pole pieces of a magnet with a uniform speed in a uniform stationary magnetic field. As the coil rotates, it cuts the lines of flux, an emf. is induced in the conductor because of relative motion between the field and the conductor. The direction of induced emf. is given by Flemings Right Hand Rule as shown in fig 1.02

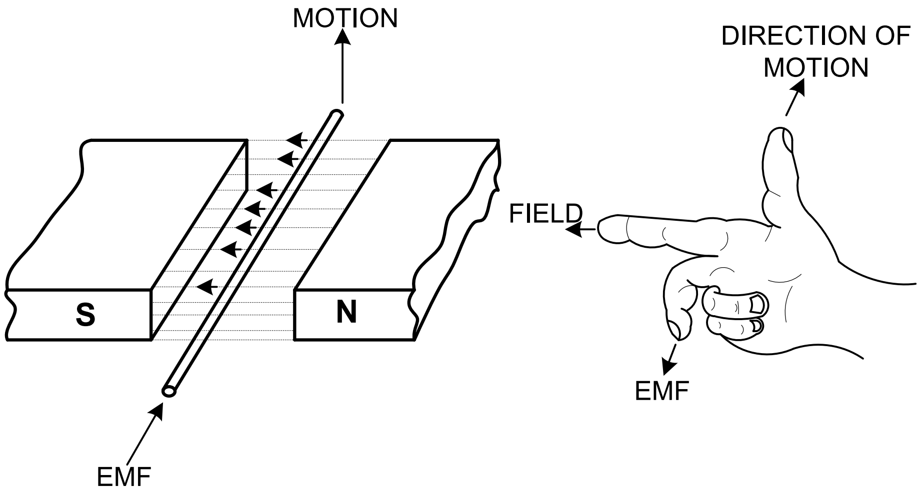

Figure 1.02 Flemings Right Hand Rule

In fig 1.03 the conductor AB which is rotating in an anticlockwise direction is at a position of cutting no flux lines, the induced emf. will be zero.

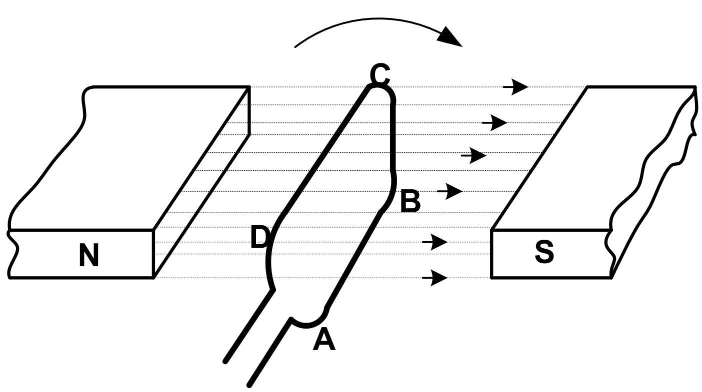

Figure 1.03 Conductor at a position cutting no magnetic lines of force

When the conductor reaches the position shown in fig.1.03 maximum e.m.f. will be induced in conductor.

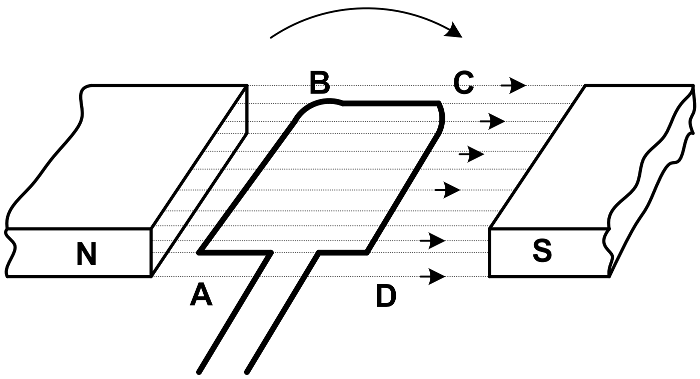

Fig 1.04 Conductor at a position cutting maximum magnetic lines of force

As the conductor moves to position shown in figure 1.04 the emf. induced in the coil is again zero since the conductor is parallel to the magnetic lines of flux.

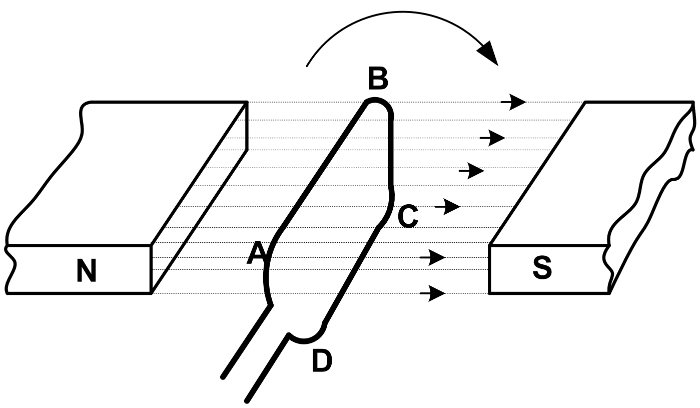

Fig 1.05 Conductor at a position cutting no magnetic lines

When the conductor AB moves to the position shown in figure 1.05, maximum emf is induced in it. The direction of the induced emf. will be found to be reversed applying Flemings Right Hand Rule.

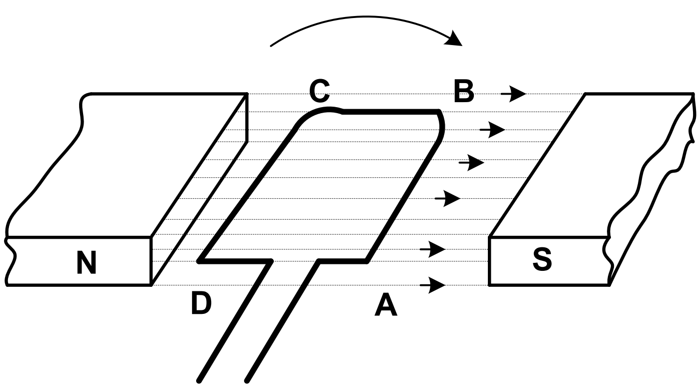

Fig 1.06. Conductor at a position cutting maximum magnetic lines of force

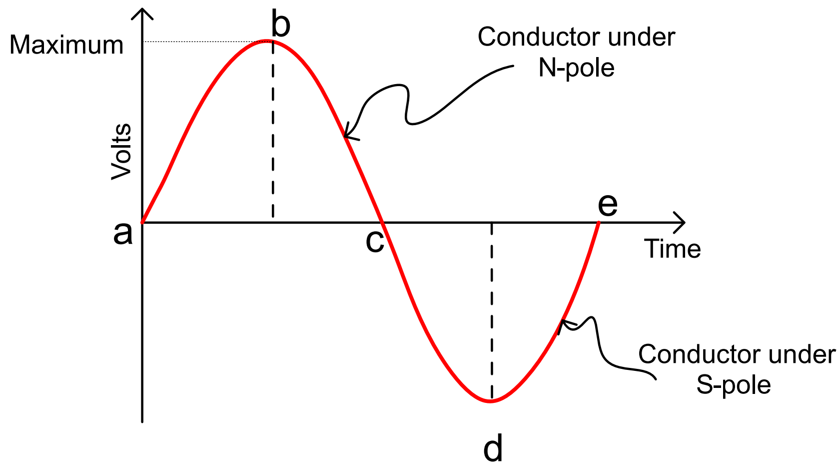

Fig 1.07. Generated emf.

The generated e.m.f.is an alternating voltage that varies in *magnitude* and *direction*.

Figure 1.06 above shows a waveform representing one complete set of changes in an alternating voltage. The figure shows the magnitude and direction of voltage in a single conductor rotating through 360º (one complete revolution) between the poles of the magnet.

Voltage variations in one revolution:

Position fig. 1 .03: The conductor is not cutting any lines of flux and therefore the voltage is zero.

Position fig. 1.04: The conductor is cutting lines of flux at right angles and therefore producing maximum voltage.

After *position fig.1.05*, the conductor is cutting lines of flux in the opposite direction and therefore the voltage is in the opposite direction.

The generated voltage is a varying quantity that generates a sign wave as shown below.

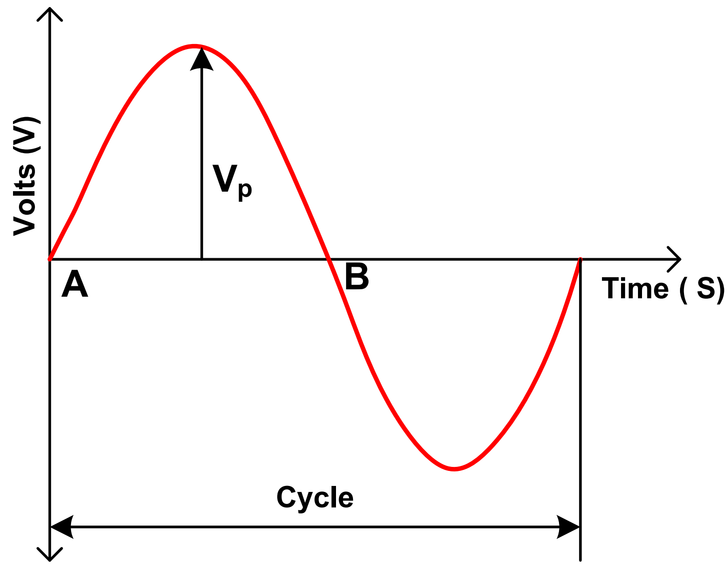

Figure 1.08 sine wave

Terms associated with the sine wave

Cycle

A cycle is one complete set of changes that are repetitive. In figure 1.07 the cycle starts from A to C.

Instantaneous value

This is the value of the alternating voltage at any given time.

Peak value

This is the maximum value of the alternating voltage

Average value

This is the arithmetical average of an alternating quantity over one cycle.

Root Mean Square (R.M.S.) value

This is the value of alternating voltage or current that will give the same heating effect as a steady voltage or current (D.C.) when passing through the same resistance.